Turbulence through canonical flows

Transition, spectra, examples, data, and computation

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IBFE Tutorials

Guiding idea

Canonical flows isolate mechanisms that recur in more complicated systems.

Adapted from a lecture by Steven L. Brunton.

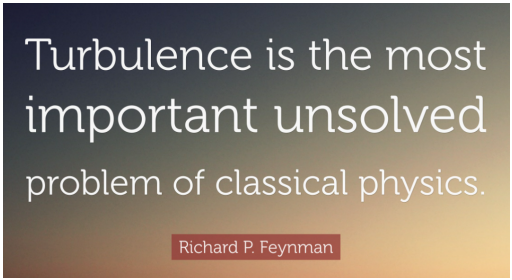
AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

Lecture roadmap

- Why turbulence is hard (and why it matters)
- Canonical flows: a "vocabulary" for turbulence
- Examples: pipe, wake, boundary layer, cavity, jets, mixing layers
- "More complex" flows (geometry, separation, multi-scale physics)
- Open-source resources: databases and CFD codes

Why turbulence is hard

- Turbulence is a *multiscale*, nonlinear dynamics problem whose observed state can depend on disturbance history.
- Small perturbations can grow, interact, and reorganize the flow.
- Many flows are *linearly stable* yet become turbulent through finite-amplitude disturbances.
- Predicting transition and statistics (drag, mixing, heat transfer, intermittency) is still challenging.



Source: to be confirmed.

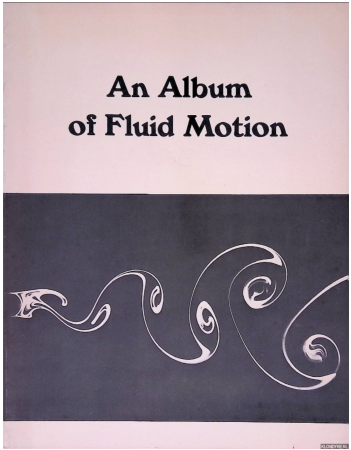
The *Feynman Lectures* (Vol. II, Ch. 41) give a classic discussion of why real viscous flows are difficult to predict.

Canonical flows (Van Dyke "album")

Idea

A small set of "canonical" configurations show up *everywhere*. Learn these and you gain a language for:

- transition mechanisms (instability $\rightarrow$ breakdown),
- coherent structures (vortices, streaks, shear layers),
- scaling laws and nondimensional groups.



M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982), cover.

A minimal tool: Reynolds number

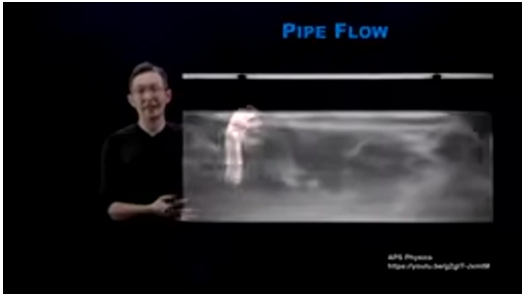
- For chosen velocity and length scales, Reynolds number compares inertia to viscosity:

$$\mathrm{Re} = \frac{UL}{\nu}.$$

- "High" Re often permits thinner shear layers and wider scale separation, but does not guarantee turbulence.
- Treat Re as a control parameter, not a universal one-to-one label for complexity: geometry, forcing, roughness, and disturbance history matter.

Pipe flow

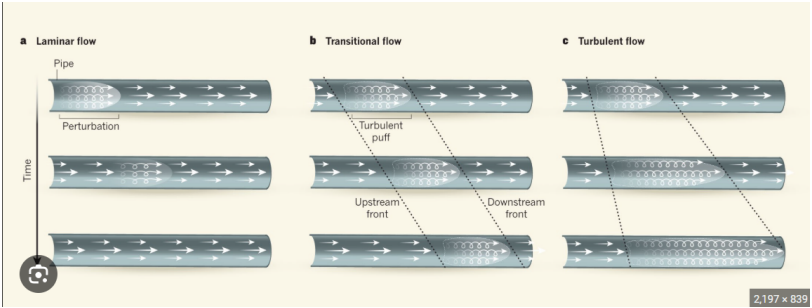
- Classic internal flow: long straight pipe, pressure-driven.
- The Hagen–Poiseuille solution exists for every Re and is linearly stable (proved for axisymmetric disturbances; computed stable to $\mathrm{Re} = 10^7$, Meseguer & Trefethen 2003), yet finite-amplitude disturbances can trigger turbulence.
- Transition often involves localized "puffs" and "slugs" (intermittency).
- Active control is possible: carefully designed perturbations can suppress turbulence (drag reduction).



Source: to be confirmed.

Pipe flow: Slugs and puffs

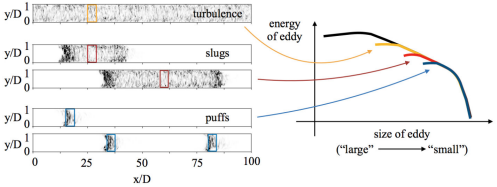
- Transitional pipe flow can coexist with laminar regions and is intrinsically intermittent.
- Localized turbulent structures have distinct dynamics:
  - **puffs** remain localized while advecting and may decay or split stochastically;
  - at higher Re, **slugs** expand by turbulent invasion at their fronts.
- The crossover is a range of behavior rather than a single universal threshold.



Source: Graham, M. Turbulence spreads like wildfire. Nature 526, 508–509 (2015).

Puffs, slugs, and spectral similarity

- The image sketches spectra sampled from the marked pipe-flow regions.
- For high-Re, locally isotropic **3D** turbulence, Kolmogorov similarity hypothesizes an inertial range
$$E(k) = C_K \varepsilon^{2/3} k^{-5/3}.$$
- This is asymptotic and statistical—not exact universality. Intermittency, anisotropy, walls, forcing, and finite Re produce deviations.
- A 2D inverse-energy cascade may also have a $-5/3$ range, but its transfer direction and mechanism differ; enstrophy transfers downscale.



Source image: Phys.org (2020).

Wake flows

- Flow past a body leaves a *wake* with a velocity deficit and strong shear.
- Bluff bodies can shed a Kármán vortex street, producing unsteady loading and noise; lifting surfaces also generate trailing and tip vortices.
- Wakes are central to:
  - aircraft wing tips and propellers,
  - wind farms and arrays,
  - animal locomotion and swimming.



Source: to be confirmed.

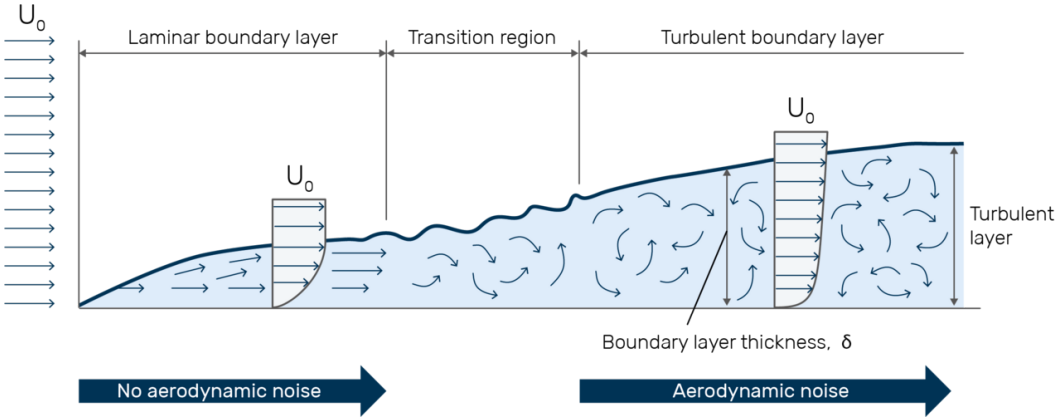
Boundary layer

- Near a wall, no-slip forces a thin region of strong shear: the boundary layer.
- Key questions:
  - when does laminar flow transition to turbulence?
  - how do near-wall streaks and vortices sustain turbulence?
  - how do roughness and pressure gradients change drag?
- Boundary layers dominate drag and heat transfer on vehicles, buildings, and animals.



Smoke visualization of a turbulent boundary layer; Van Dyke, *An Album of Fluid Motion* (turbulent boundary layer plate; also reproduced in Dou 2018).

Boundary layer: spatial development

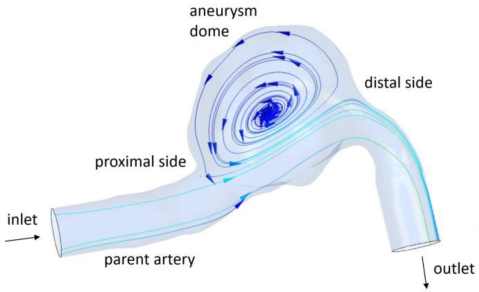


Boundary-layer schematic after GRAS Sound & Vibration, *The boundary layer—laminar and turbulent flows* (blog; source to confirm),

https://www.grasacoustics.com/blog/aerodynamic-measurements-components-of-the-boundary-layer.

Cavity flow

- In an **open engineering cavity**, a shear layer spans the opening; pressure/acoustic feedback can create resonant oscillations.
- An intracranial aneurysm is instead a **cavity-like recirculating flow**: pulsatility, geometry, and vessel-wall conditions shape its hemodynamics.
- Related applications:
  - aneurysm inflow jets and recirculation,
  - gaps, steps, and surface defects,
  - flow-induced vibration.



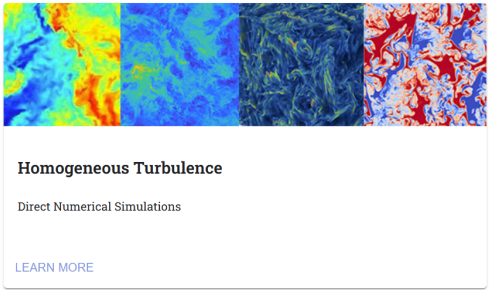
Source: Boniforti MA, Magini R, Orosco Salinas T. Hemodynamic Investigation of the Flow Diverter Treatment of Intracranial Aneurysm. Fluids. 2023; 8(7):189.

Jets, mixing layers, and separated flows (overview)

- **Round jet:** downstream of the potential core the mean profile becomes *self-similar*, the half-width grows roughly linearly with a spreading angle of order 10–12°, and Kelvin–Helmholtz roll-up of the annular shear layer seeds the initial vortex rings.
- **Plane mixing layer:** two streams of different speed produce the Brown–Roshko coherent spanwise rollers, which pair and merge; the growth rate scales with the velocity ratio $(U_1 - U_2)/(U_1 + U_2)$ rather than with Re.
- **Separated flows:** a backward-facing step separates at a fixed geometric edge; at high Re the mean reattachment length is about 6–7 step heights, and the shear layer, recirculation bubble, and reattachment zone each have their own statistics.
- **Combinations:** real configurations superpose these—a jet issuing into a cavity, a wake feeding a mixing layer, a step downstream of a boundary layer.
- **Multi-scale physics:** geometry sets the large scales; viscosity sets the small ones. The gap between them is what makes prediction expensive.

Open data: turbulence databases

- Public datasets let you test ideas without running a massive DNS yourself.
- Example: Johns Hopkins Turbulence Databases (JHTDB)
  - isotropic turbulence, channel flow, boundary layers, geophysical, ...
  - query-able via web/API-style interfaces
  - great for teaching statistics, structure functions, spectra, and coherent-structure detection



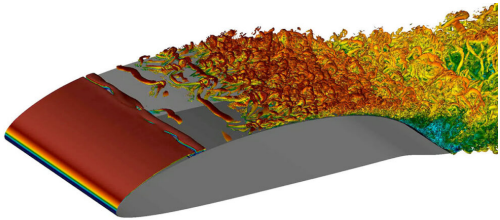
Johns Hopkins Turbulence Databases, turbulence.pha.jhu.edu.

Open-source CFD codes

Two common ecosystems

- **OpenFOAM**: open-source finite-volume framework with many RANS/LES solvers.
- **Nek5000**: high-order spectral-element code; widely used for DNS/LES in complex geometries.

- Choosing a tool depends on: accuracy needs, geometry, turbulence model, and HPC resources.



Nek5000 project banner image, Argonne National Laboratory, nek5000.mcs.anl.gov.

Takeaways

- Turbulence is everywhere, but canonical flows give structure to the chaos.
- Each canonical flow highlights different mechanisms (instability, shear, walls, separation, resonance).
- Open videos, datasets, and codes make it easier than ever to *see* turbulence and to explore it quantitatively.

Selected references and source notes

- R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, Vol. II, Ch. 41, "The Flow of Wet Water".
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- A. N. Kolmogorov, "The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers," *Proc. R. Soc. A* **434** (1991 translation), 9–13. doi:10.1098/rspa.1991.0075.
- R. H. Kraichnan, "Inertial ranges in two-dimensional turbulence," *Phys. Fluids* **10** (1967), 1417–1423. doi:10.1063/1.1762301.
- H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed., Springer (2017).
- Official resources: JHTDB, OpenFOAM Foundation, and Nek5000.